Relative Sizing

Requirements can be estimated in terms of their size and the amount of time it will take to complete them. It is common for the estimate to be added to each user story on a product backlog when using an agile approach. There are many different methods of relative sizing, including T-shirt sizing (small, medium and large), or story point estimation which considers the amount of effort required in relative terms.

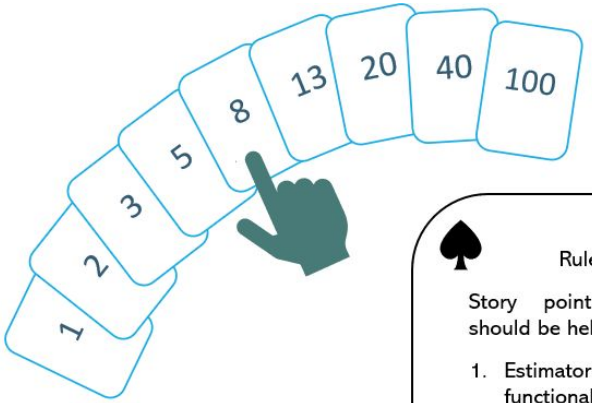
Sizing is necessary when using an agile approach because it is a prerequisite of sprint planning. It is less common to add sizes to requirements when using waterfall (although the project manager and IT lead will no doubt discuss development effort).

Planning Poker

Story point estimation considers the amount of effort required in relative terms. Estimations of effort are provided by the development team, testers and sometimes the BA. Project Planning Poker cards are often used.

The Poker cards start off in the **Fibonacci** number sequence. First 1, then 2, then add the previous number to the one before it to get the next number, so the sequence continues with 3 (1+2), then 5 (3+2) etc. To keep things simple, 21 has been replaced with 20 and the larger stories can only be sized with 40 and 100. It is likely that a story estimated as 100 story points is an epic and needs to be decomposed into smaller stories and estimated again.

Planning Poker



The team reveal their estimation efforts to each other. If their estimates are the same or very similar they are averaged. Outliers are discussed and the game is replayed, this usually brings estimations closer.

Rules of the Game

Story point estimation sessions should be held regularly.

1. Estimators should be a cross functional group. Each Estimator holds a set of Planning Poker cards.
2. The product owner reads the product backlog item.
3. The item is discussed.
4. Each estimator selects the card that represents their estimation.
5. All estimators reveal their cards
6. If all cards are the same, the estimate is documented. If all cards are not the same, the group discusses the estimates, focusing on the outlying values.
7. Repeat until all estimates are the same, or similar enough to take an average.

Planning Poker

Note that a story point does not necessarily relate to a period of time. When an agile team first forms, they may use a rough guide, such as 1 point = 1 day of team effort, however over time this may change or be recalibrated. An experienced team instinctively knows what a story point means in terms of time/effort and can size user stories relative to each other as well as understanding how many stories of varying sizes the team can develop within the sprint.

Key Points about User Stories

Collaboration	•**User stories grow in collaborative environments.** The workshop environment provides a great forum for agreeing user stories. This allows the stakeholders to work closely together to agree on requirements and their priorities.
User-centric	•**User stories are user-centred.** Always ensure your user stories are focused on the user, and that they're simple and concise.
Top down	•**User stories may come from epics.** One method is **to s**tart with epics and go into more detail later to produce the underlying user stories after more detailed conversations have taken place.
Confirmations	•**User stories must include acceptance criteria.** This ensures that the story describes a feature that is testable and it pinpoints the users expectations of the goals of the feature.
Estimated	•**The amount of effort required to develop and test the story should be known.** This is usually added in a measurement known as story points where the relative effort is understood, as is the number of story points that each team can develop in a sprint.
Transparency	•**User stories are most effective in an environment of transparency.** One way to go about this is to write them on cards and display on team boards, so everyone can see what the requirements are, and their progress in development.

Velocity

The amount of work that each team can carry out within a sprint is usually determined by how much work was done in previous sprints. Using previous experience is often referred to as yesterday's weather.

The expectations, or forecast, is known as the team's velocity, for example, they may have a velocity of 24 story points.

However, we all know that the weather changes and estimates can be inaccurate and the climate changes over time. To cope with change, teams recalibrate their velocity frequently, so that they are continuously working at an optimum level. Not overstretched, and not running out of tasks to do (which can lead to loss of momentum).

Scrum

- Scrum is a popular agile framework
- Developed by Schwaber and Sutherland
- Based on the Three Pillars of Empiricism

Click here to find out more about Scrum,

Kanban

Kanban offers supplementary techniques to use with any Agile methodology. It is based on six core practices.

1. Visualise the work
2. Limit work-in-progress
3. Make policies explicit
4. Measure and Manage flow
5. Implement feedback loops
6. Improve collaboratively, evolve experimentally

Click here if you would like to know more about Kanban.

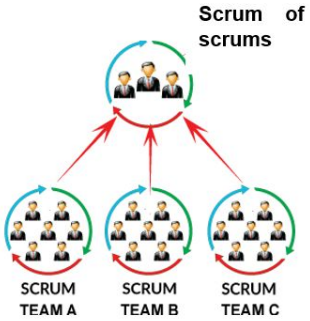
Scaled Agile

There are many Agile methods, such as Scrum of Scrums, SAFe, LeSS and Scrum@Sale, which are known as scaled Agile. They are often used to facilitate larger more complex business change projects.

LeSS Framework

SCRUM @SCALE

SAFe® PROVIDED BY SCALED AGILE®



Scaled Agile

There are some other practices to try first to remove the need for scaling.

- Decomposing into independent goals
- Shortening cycle time
- Refactoring the solution architecture
- Decoupling team dependencies